

प्रश्न क्र.
Q.No.

(Weightage: 05 marks)

2 marks

- ① List types of Operating System.
- ② Define any Operating System.
State its application.
- ③ Structure of OS Draw.
- ④ State two features of Linux.
- ⑤ Advantages & DisAdvantages of any operating system.

4 marks

- ① Explain multiprogramming o.s.
or
- ② multiprocessor & Advantages.
or
- ③ Time sharing with Diagram
- ④ Real time & its types.

Comparison

- ① CLI vs GUI (4m/6m)
- ② Time sharing vs Real Time (2m/4m)
- ③ Time Sharing (multitasking) vs multiprogramming (2m/4m)

Expected

- ① Evolution of Operating System (5m)
- ② iOS vs Android (4m/2m)
- ③ Components of Computer System (4m)
- ④ List Operations on O.S (2m)

Unit II
(Weightage: 10 marks)

2marks

- ① List services of Operating System
or
List System calls.
- ② List down system calls of file management.

4marks

- ③ Explain any four services of Operating System.
- ② What is purpose of System call? Explain system calls and its types.
- Imp ③ Components of Operating System.

6marks

- ① Use of operating system tools:
 i) User management
 ii) Security policy
 iii) Device Management
 iv) Task Scheduler
 v) Performance monitor

प्रश्न क्र.
Q.No.

Unit III

(weightage : 14 marks)

2 marks

- ① Draw Process state diagram.
- ② Draw PCB Block.

4 marks

- ① List scheduler types & explain each scheduler.
- ② Explain the concept of CPU switching and context switching.
- ③ Explain User Thread and Kernel Thread with Advantages and Disadvantages.
- ④ Explain Process state Diagram.
- ⑤ Draw and Explain Process Control Block.
- ⑥ Explain commands with syntax
i) PS ii) wait iii) sleep
iv) exit v) kill
- ⑦ Discuss the Benefits of multithread programming.

6 marks

- ① Explain Interprocess communication.
1) Shared memory 2) message passing.
- ② Explain & draw Multithreading model.

प्रश्न क्र.
Q.No.Competition

- ① Shared memory VS Message passing
 ② Process VS Thread
 ③ Process VS Program

Expected

Describe method of executing given process command. (2m/4m)

Commands2marks

- ① Give Commands to perform following tasks:
- To add delay in script.
 - To terminate a process.

② 4marks

- Create folder OSY
- Create file First in OSY folder.
- List/display all files & directories.
- Write command to clear screen.

6marks

- wait 238501
- sleep 09
- PS - u Asha

Output

Unit IV

(weightage: 14 marks)

2 marks

- ① Define CPU and I/O burst cycle.
- ② Define CPU burst program
I/O burst program

4 marks

- ① Explain Round Robin with example.
- ② Explain CPU & I/O cycle with example.
- ③ Explain four Scheduling Criteria.
- ④ Define Deadlock & necessary conditions for deadlock.

OR

Define Deadlock & preventive conditions for deadlock.

- ⑤ State steps for Banker's algorithm.

- ⑥ Explain Pre-emptive, is better than Non-Pre-emptive.

6 marks

- ① Numerical (FCFS, STF, Round Robin)
- ② Explain Multilevel Queue with example.

Comparison

- ① Pre-emptive vs Non-Pre-emptive (2/4)

Expected

- ① Define : Pre-emptive & Non-Pre-emptive (2 marks)
- ② What is CPU scheduling & (2 marks)
- ③ Need of Scheduling (2 marks)

Unit II

(weightage: 14m)

2 marks

- ① Define
- i) Virtual memory
 - ii) Paging
 - iii) Segmentation
 - iv) Fragmentation
- ④ Page fault & Page hit

- ② List types of Free space management techniques.

4 marks

- ① Types of partitioning & Explain
- ② variable partitioning with suitable example.
- ③ Explain free space management technique enlist all and explain any one.
- ④ Define Fragmentation. Explain Internal and External Fragmentation.

6 marks

- ① Numerical (FIFO, LRU, OPTIMAL)
- ② List and Explain Free space management techniques in detailed.
- ③ Describe concept of virtual memory with respect to paging. Also draw paging hardware diagram and describe its working with example.

Comparison

Paging vs Segmentation

Expected

- ① Variable vs Static Partitioning
- ② Define LRU
- ③ Define
 - i) Swapping
 - ii) Dynamic ReAllocation.

प्रश्न क्र.
Q.No.

Unit - VI

(weightage 10m)

2 marks

- ① Types or examples of file
- ② File attributes.
- ③ Operation on files.

6 marks

- ① Explain File directory structure.
 - i) Single-level
 - ii) two-level
 - iii) tree-level.

4 marks

- ① File allocation methods?
- ② Explain linked allocation method?
- ③ Access methods :-
 - 1) Sequential
 - 2) Direct
 - 3) Swapping

Do not write your name or seat no. below this line

Q.No.						Q.No.					
-------	--	--	--	--	--	-------	--	--	--	--	--

Q.No.

⑤ Best Fit
Worst Fit
First Fit (numerical)

End

